



BACKTRACKING -

Subset Sum

EXAMPLE

Given $n + 1$ positive numbers: w_i , $1 \leq i \leq n$ and M , find all subsets of w_i whose sum is M .

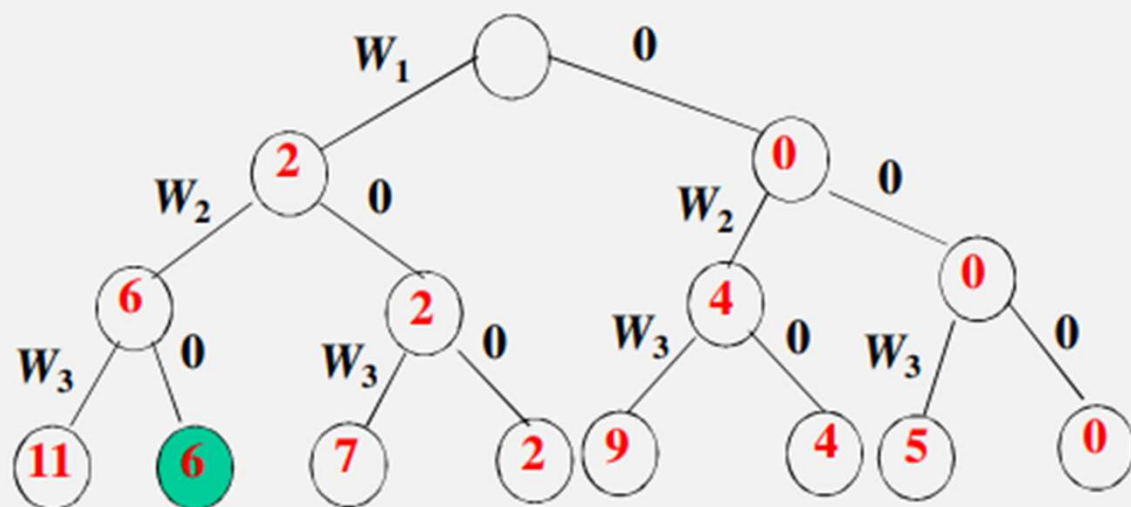
- For ex. If $n = 4$, $(w_1, w_2, w_3, w_4) = (11, 13, 24, 7)$ and $M = 31$, then the desirable subset are $(11, 13, 7)$ and $(24, 7)$.
- If represented in terms of indices, then the solution is $(1, 2, 4)$ and $(3, 4)$.
- In general the desired solution is k -tuples $(x_1, x_2, \dots, x_k)$ $1 \leq i \leq k$ and different solutions may have different size tuples.

Example 2

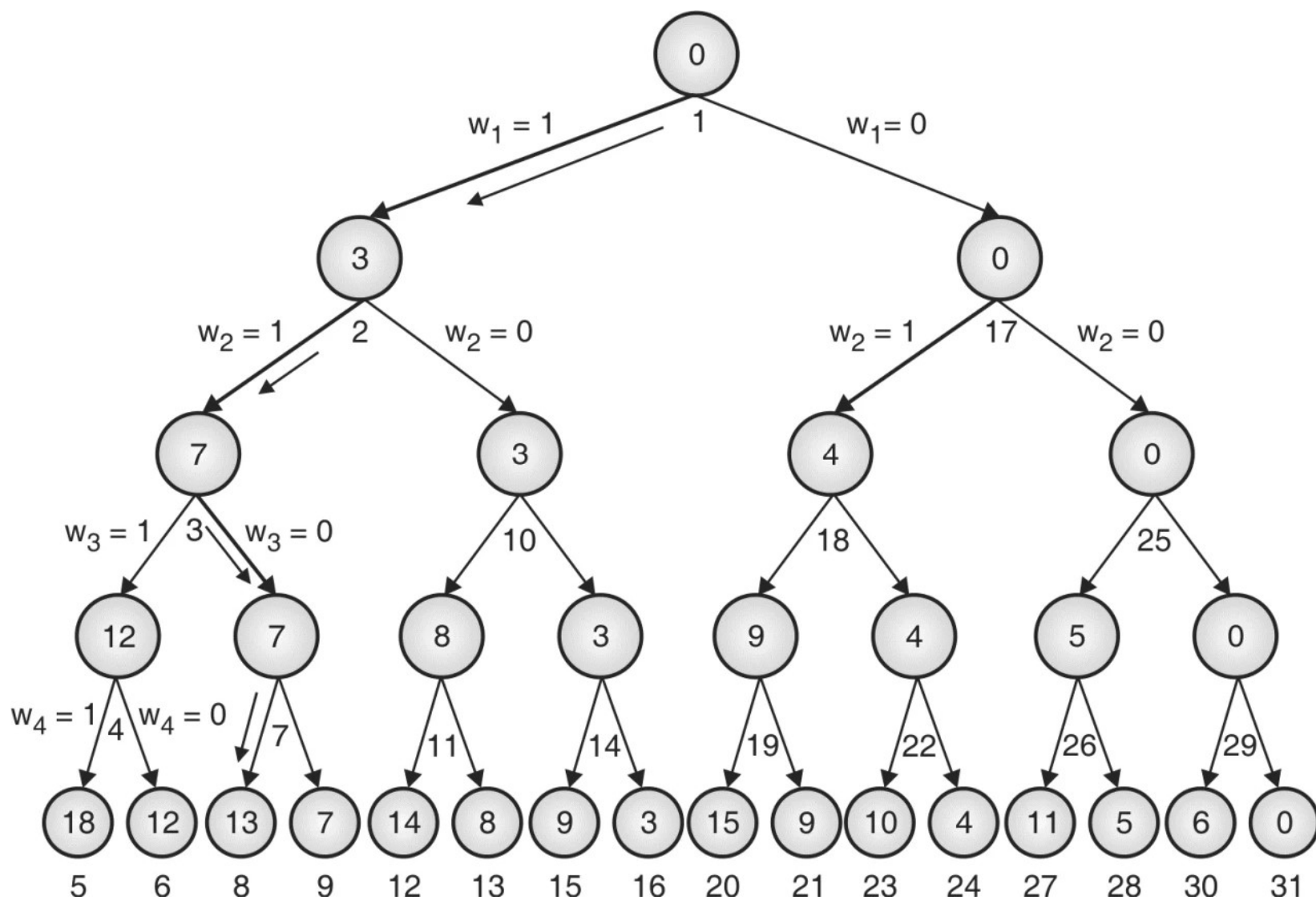
Suppose that $n = 3$, $M = 6$, and

$$w_1 = 2, w_2 = 4, w_3 = 5.$$

Find the solutions.



- Consider the sum-of-subset problem, $n = 4$, $\text{Sum} = 13$, and $w_1 = 3$, $w_2 = 4$, $w_3 = 5$ and $w_4 = 6$. Find a solution to the problem using backtracking. Show the state-space tree leading to the solution.



- $W = (5, 10, 12, 13, 15, 18)$; $M = 30$